Customer Segmentation and Data-driven Sales Strategy for Turtle Games

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Background

- Turtle Games wants to improve overall sales performance by understanding and utilising customer trends.
- Turtle Games needs a data-informed approach to customer segmentation, prioritisation, sentiment analysis and to using loyalty points data for predicting and informing sales performance.
- Findings aim to support targeted interventions to understand and improve sales.

Business questions

- What factors influence loyalty point accumulation?
- Can customers be grouped into meaningful segments based on age, income (remuneration), or spending habits?
- How do customers feel about products and experiences based on their reviews and feedback?
- Can we leverage the loyalty points data to develop descriptive statistics and provide insights into where should the business focus its efforts for maximum impact?

Analytical approach

- Assess current loyalty point accumulation and it's relationship with customer age, remuneration, education, gender and spending habits.
- Identify the parameters that have the strongest impact on loyalty point accumulation to drive customer segmentation.
- Analysing customer reviews and feedback to assess customer sentiment and it's variance across segments.
- Predicting loyalty point accumulation based on relevant customer data to inform marketing approach.

Analytical approach

Python was used for:

- Data cleaning
- Customer segmentation (clustering)
- Sentiment analysis of reviews

R was used for:

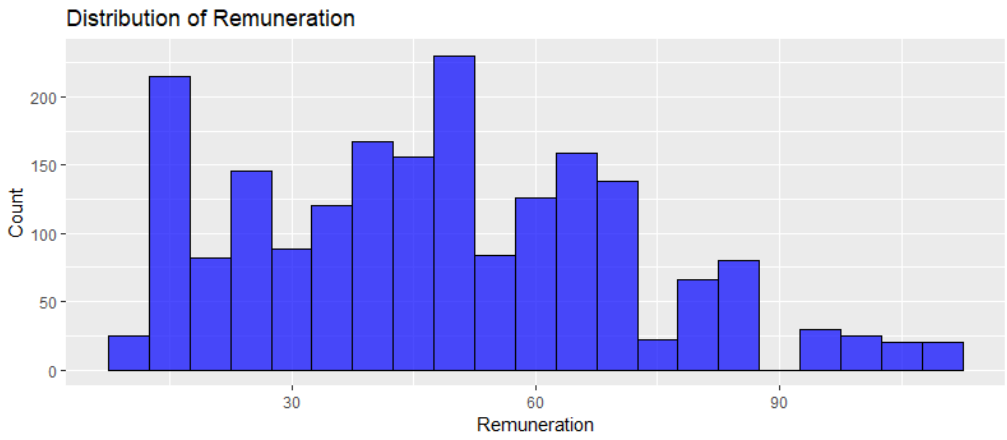
- Exploratory data analysis
- Statistical modelling to understand loyalty drivers

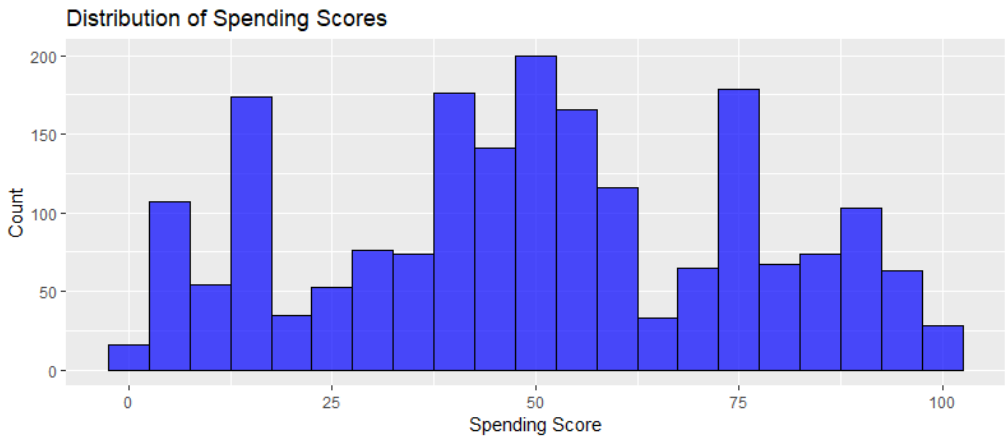
Process overview:

- Cleaned and validated customer and review data
- Explored patterns using charts and summary statistics
- Built models to explain loyalty behaviour
- Grouped customers into clear segments
- Analysed customer sentiment to add qualitative insight

Demographic data at a glance

Wide range in customer demographics confirms opportunity for segmentation

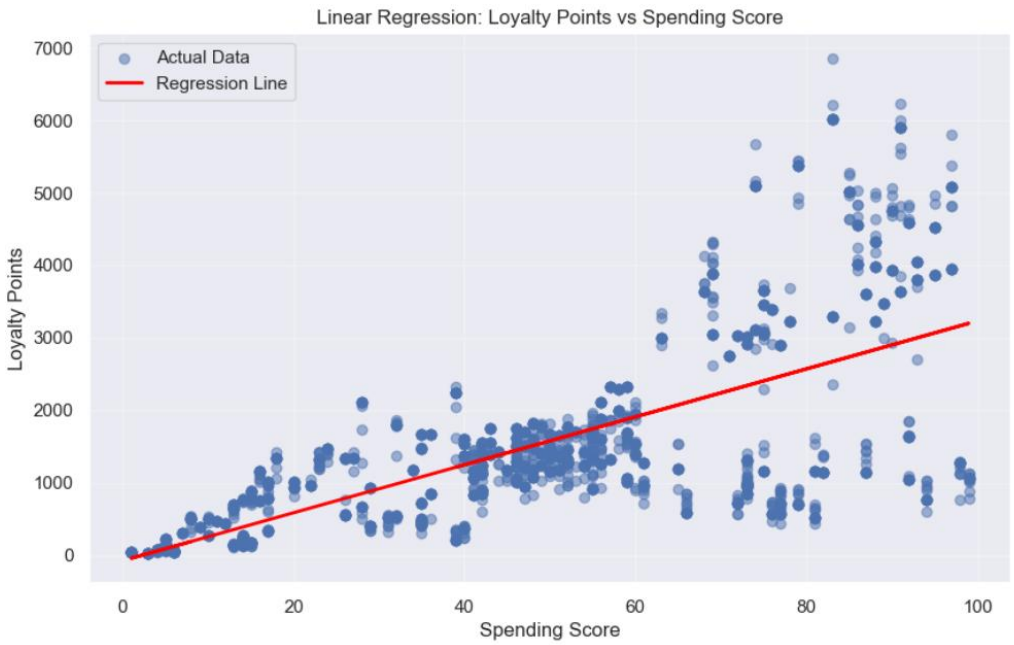




- 2,000 entries with consistent, complete data
- Data shows a wide range of income and spending levels with customer age ranging from 17-72 years.
- Loyalty point accumulation varies significantly across customers.
- Customer segmentation is essential for driving optimal sales growth.

Key insights from Regression

Spending score & remuneration are the strongest predictor of loyalty points
Age plays a smaller but still positive role



- Model explains ~45.2% of the variance.
- **One-unit increase in spending score** is associated with an increase of **~33 loyalty points**

- Model explains ~38% of the variance.
- **One-unit increase in remuneration** is associated with an increase of **~34 loyalty points**

Key insights - Multiple Linear Regression

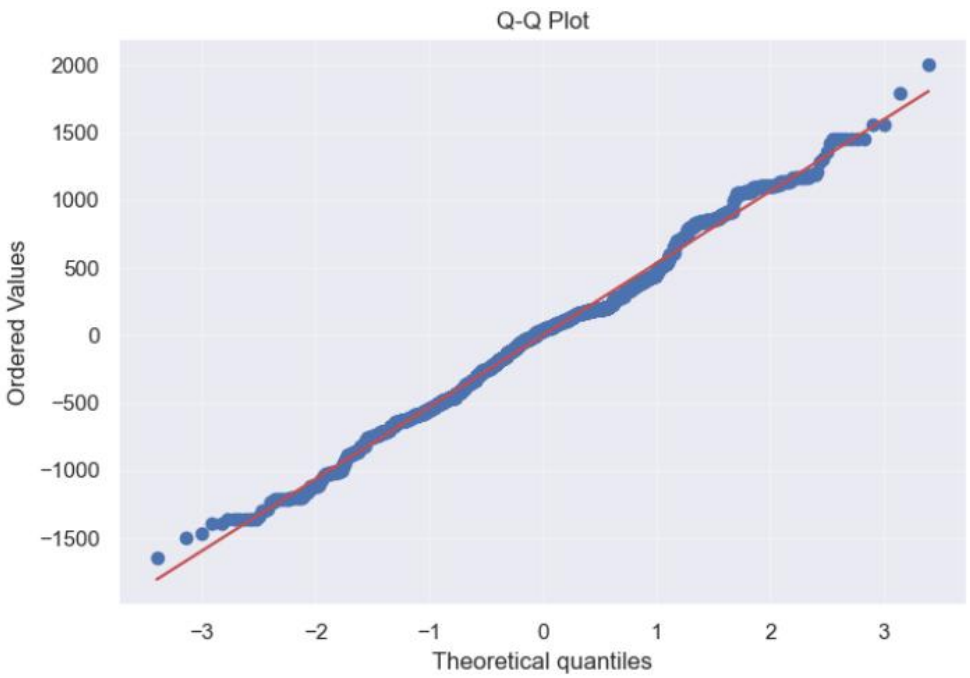
Spending score & remuneration taken together enable a stronger predictive model of loyalty points

```
                            OLS Regression Results
==============================================================================
Dep. Variable:         loyalty_points   R-squared:                       0.827
Model:                            OLS   Adj. R-squared:                  0.827
Method:                 Least Squares   F-statistic:                     4770.
Date:                Sun, 11 Jan 2026   Prob (F-statistic):               0.00
Time:                        20:02:03   Log-Likelihood:                -15398.
No. Observations:                2000   AIC:                         3.080e+04
Df Residuals:                    1997   BIC:                         3.082e+04
Df Model:                           2
Covariance Type:            nonrobust
==================================================================================
                     coef    std err          t      P>|t|      [0.025      0.975]
----------------------------------------------------------------------------------
const          -1700.3051     35.740    -47.575      0.000   -1770.396   -1630.214
spending_score    32.8927      0.458     71.845      0.000      31.995      33.791
remuneration      33.9795      0.517     65.769      0.000      32.966      34.993
==============================================================================
Omnibus:                        4.723   Durbin-Watson:                   3.477
Prob(Omnibus):                  0.094   Jarque-Bera (JB):                4.650
Skew:                           0.103   Prob(JB):                       0.0978
Kurtosis:                       3.115   Cond. No.                         220.
==============================================================================
```

- When spending score and remuneration are taken together, the model explains ~83% of the variance in loyalty points.
- Both predictors and f-statistic are highly significant.
- This indicates suitability of this model to predict loyalty points based on spending behaviour and remuneration. Age is seen to play only a small to negligible role.
- **These predictions can help Turtle Games estimate loyalty points for new customers.**

Key insights

MLR-based model suitably predicts loyalty point accumulation

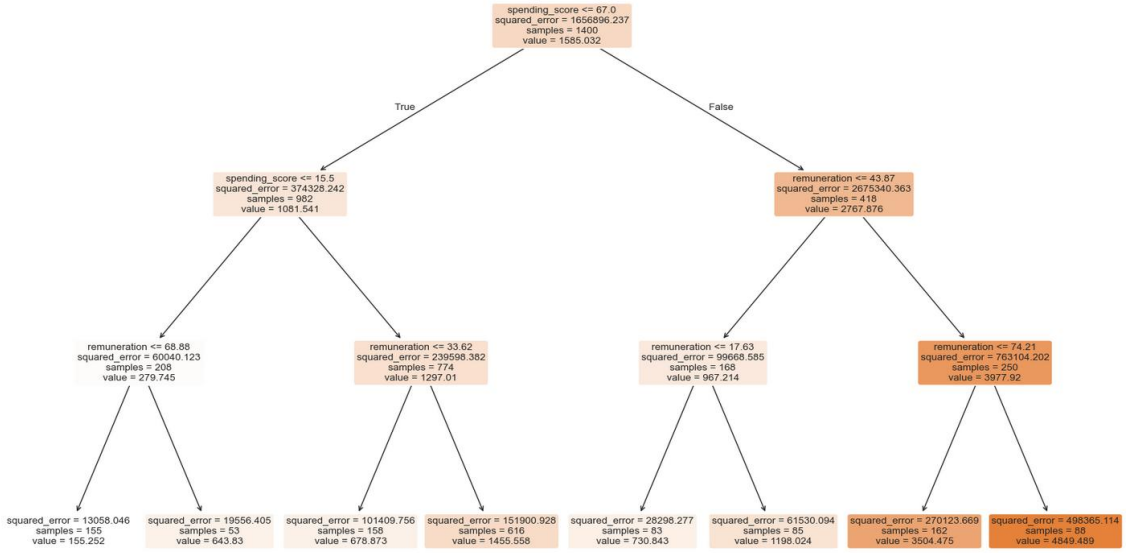



```
--- Prediction Scenarios ---
   spending_score  remuneration  predicted_loyalty_points
0              25            25                 -28.500259
1              50            50                1643.304578
2              75            75                3315.109416
3              90           100                4657.987307
```

- **The Q-Q plot suggests only minor deviations** between observed and predicted values from normality and predominantly only in extreme cases.
- **The model predicts high loyalty points for High-spend, high-income customers as also seen in the actual dataset.**
- These findings suggest that loyalty strategy should **prioritise spending-based and remuneration-based segmentation** over demographic targeting.

Decision trees

Spending score & remuneration verified to be key predictors of loyalty



- Decision tree regressors are an alternative predictive model that does not assume linearity.
- Decision tree is pruned to consider only top 3 dominant features.
- After pruning the tree, the model retains strong explanatory power ($R^2 \approx 0.91$), indicating that the key drivers of loyalty points were preserved.

Customer segmentation

Majority of the customers (38.7%) fall under the "Balanced shoppers" cluster
"Affluent conservatives" cluster represents a high-value, untapped segment.

- Five distinct customer types observed from k-means clustering
- These clusters are labelled based on their income and spending patterns.
- These insights allow the marketing department to design tailored campaigns.

Customer segmentation

Each cluster requires a different engagement strategy.

Five clusters are identified based on remuneration and spending score. The recommended actions and customer profiles are as tabulated below.

Cluster Name	Remuneration	Spending Behaviour	Customer Profile	Business Opportunity	Recommended Actions
Budget Essentials	Low	Low	Cautious, necessity-driven customers with potentially high price sensitivity	Stable but low-margin segment	Promotions on essentials, discount bundles, value packs
Aspiring Spenders	Low–Medium	High	Customers spending disproportionately relative to income	High growth and loyalty potential	Rewards programs, BNPL options, targeted promotional offers
Balanced Shoppers	Medium	Medium	Core customer base with predictable, value-oriented purchasing behaviour	Reliable revenue contributors	Retention campaigns, cross-selling, personalised recommendations
Premium Enthusiasts	High	High	High-value customers with strong purchasing power and loyalty potential	Key profit-driving segment	VIP programs, exclusive offers, early product access
Affluent Conservatives	High	Low	Financially capable customers with conservative spending habits	Untapped revenue opportunity	Product education, premium trials, targeted nudges

Sentiment analysis

Reviews are generally positive

game
play
one
great
fun
card
time
make
love
player
well
new
different
expansion
little
playing
book
played
take
piece
good
many
even
find
first
rule
use
much
work
people
version
think
box
set
help
buy
kid
gift
two
enough
friend
way
used
nice
thing
add
bought
feel
point
shape
monster
loved
quality
puzzle
price
year
old
product
another
hard
toy
hour
great game
small
seem
children
board
game
still
dd
really
back
keep
robot
always
made
come
turn
give
go
ball
bit
group
know
easy
day
board
hand
dungeon
together
got
need
using
instruction
adult
stock
place
every
probably
acquire
plastic
recommend
miniature
everything
learn
ball whacks
fun
getting
around
though
child
want
already
sticker
looking
found
say
better
making
number
excellent
family
strategy
age
difficult
review
yarn
fit
try
able
gets
far
game play
others
draw
quest
item
fan
quite
waterdeep
lords
mini
best
hold
magnet
animal
daughter
definitely
perfect
look
without
part
everyone
upon
purchased
cardboard
actually
see
fun game
getting
cute
interesting
color
easily
favorite
three
might
acquire
building
must
mini
place
stock
every
probably
hand
board
day
able
try

Business recommendations

1. Prioritise High-Value Customer Segments with Tailored Engagement

(High Priority | Short-term: 0–6 months)

Why:
Clustering analysis identified a distinct **“Premium Enthusiasts”** segment with both high remuneration and high spending. Retaining and deepening engagement with this group offers the fastest return on investment.

Actions:

- Introduce VIP loyalty tiers with exclusive rewards and early product access.
- Personalise marketing campaigns using spending and sentiment signals.
- Offer premium bundles or limited-edition products to reinforce exclusivity.

KPIs:

- 5–10% increase in average loyalty points per Premium customer within 6 months.
- Improvement in repeat purchase rate for this segment.
- Positive uplift in sentiment scores for premium-targeted communications.

Business recommendations

2. Activate Underperforming but High-Potential Customers

(Medium Priority | Medium-term: 6–12 months)

Why:
The **"Affluent Conservatives"** show high remuneration but relatively low spending, indicating untapped revenue potential.

Actions:

- Deploy targeted education and trial campaigns to reduce perceived purchase risk.
- Use personalised incentives (e.g. first-purchase bonuses, premium trials).
- Highlight product value and long-term benefits in communications.

KPIs:

- Increase average spending score of this cluster by 10% within 12 months.
- Growth in total loyalty points generated by this segment.
- Conversion rate from low to medium spending behaviour.

Business recommendations

3. Protect and Retain the Core Customer Base

(High Priority | Ongoing)

Why:
“Balanced Shoppers” form the largest proportion of the customer base and provides stable, predictable revenue. While individually less valuable than premium customers, the business’s long-term performance depends heavily on keeping this group engaged.

Actions:

- Maintain consistent loyalty rewards and retention-focused campaigns.
- Cross-sell complementary products based on past purchasing patterns.
- Monitor sentiment trends to proactively address dissatisfaction.

KPIs:

- Stable or increasing retention rate within the Balanced Shoppers segment.
- Sustained contribution to total loyalty points.
- Reduction in negative sentiment signals within this cluster.

Conclusions and next steps

- Turtle Games' customer base is diverse.
- Spending behaviour along with remuneration are the strongest drivers of loyalty.
- Customer segmentation along spending score and remuneration enables smarter targeting, better retention, and higher lifetime value.
- Overall sentiment is generally positive
- Loyalty point accumulation and customer segmentation can together be used for defining sales and marketing strategies using predictive models.

Next steps:

- Repeat the analysis in 6 months time following implementation of interventions
- Include more features in data collection such as product categories, customer tenure, basket sizes and customer ID for frequency analysis

Thank you